

CS5239 Computer System Performance Analysis

L4: Markovian Queueing Models

Markovian Queueing Models

- Markov Models
- General Birth-Death Process Queueing Models
 - M/M/1 Queueing System
 - M/M/m Queueing System
 - M/M/m/K Queueing System
- Queueing networks
- An example: reliability of a data center
- Summary



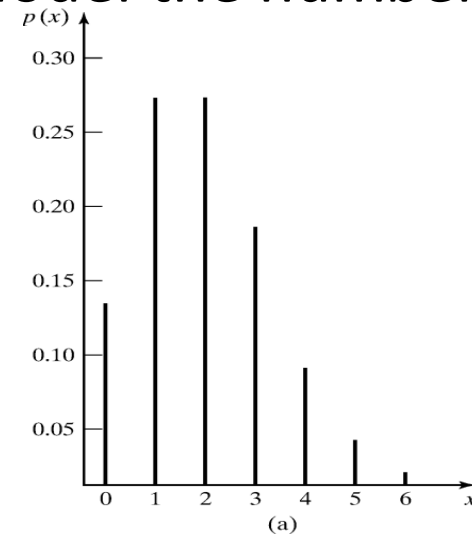
Andrei Andreyevich **Markov**

Markov Models

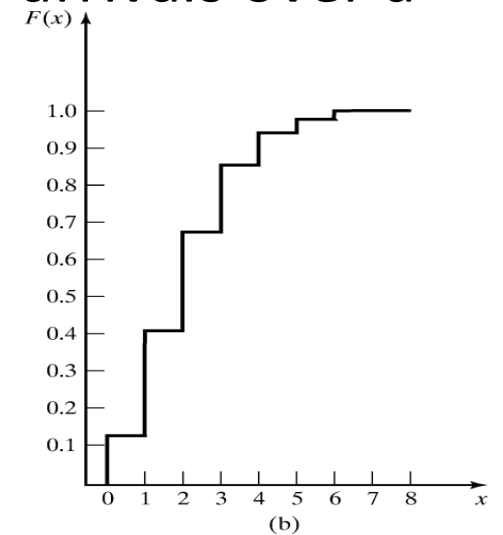
- Markov chains
 - Are stochastic models describing a sequence of possible events where the probability of each event depends only on the most recently known state
 - Memoryless property
 - In a continuous-time Markov chain, the transition from one state to another is characterized by an exponential distribution
 - The exponential distribution is the only continuous distribution that has the memoryless property

Poisson Distribution: Poisson (λ) (from Lstats)

- Commonly used in queuing models to model the number of arrivals over a given interval
- Key characteristics
 - Parameters:
 - λ is the mean number of events within a time interval, i.e., $\lambda = \text{mean}$, $\lambda > 0$
 - Range: $x = 0, 1, 2, \dots, \infty$
 - Mean and variance are both equal to λ



Probability mass function
(shows the probability of x
events in the same interval)

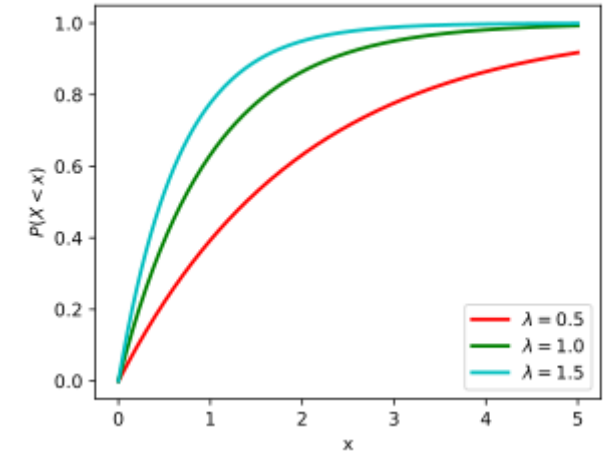


Cumulative distribution function
(shows the probability that X
will take a value $\leq x$)

<https://youtu.be/rBIQmwaoZfs?t=24>

Exponential Distribution

- Properties of the exponential distribution
 - $F_X(x) = P[X \leq x] = 1 - e^{-\lambda x}$ - **cumulative distribution function**
 - Average and standard deviation of an exponentially distributed random variable with parameter λ are both equal to $1/\lambda$. Thus, the coefficient of variation is equal to 1
- Poisson arrivals imply that inter-arrival times are exponentially distributed
 - If inter-arrival times are exponentially distributed with parameter λ then the probability of exactly n arrivals in a given time period from time 0 to time t is
 - $P[n \text{ arrivals in } (0,t)] = \frac{e^{-\lambda t} (\lambda t)^n}{n!}$



Example: Poisson (λ)

The number of visitors to a webserver per minute follows a Poisson distribution. If the average number of visitors per minute is 4, what is the probability that

- i. there are two or fewer visitors in one minute?
- ii. there are exactly two visitors in 30 seconds?

Example: Poisson (λ)

i. If the average number of visitors in a minute is 4, $\lambda = 4$, what is the probability that there are two or fewer visitors in one minute?

Calculate $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$

$$P(X = 0) = \frac{e^{-4}4^0}{0!} = e^{-4}$$

$$P(X = 1) = \frac{e^{-4}4^1}{1!} = 4e^{-4}$$

$$P(X = 2) = \frac{e^{-4}4^2}{2!} = 8e^{-4}$$

Thus, the probability of two or fewer visitors in a minute is $e^{-4} + 4e^{-4} + 8e^{-4} = 0.238$

Example: Poisson (λ)

ii. If the average number of visitors in 1 minute is 4, what is the probability that there are exactly two visitors in 30 seconds?

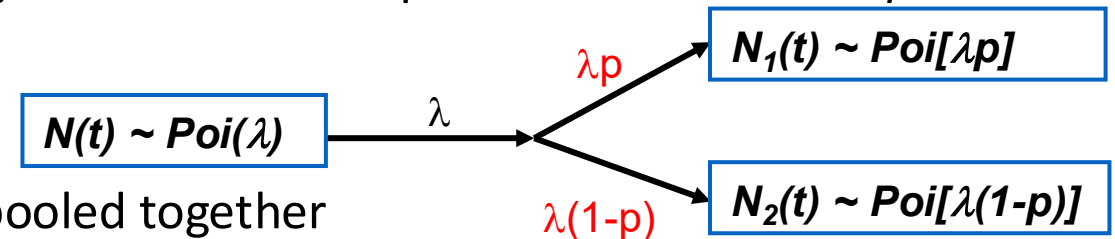
Thus, $\lambda = 2$ and
$$P(X = 2) = \frac{e^{-2}2^2}{2!} = 2e^{-2} = 0.271$$

Splitting and Pooling

Consider a Poisson process $\{N(t), t \geq 0\}$ with (arrival) rate of λ

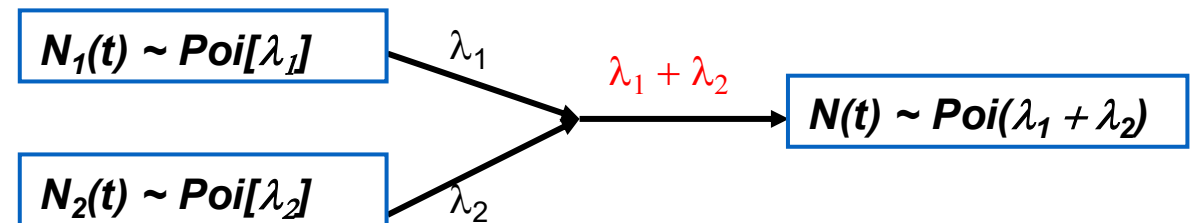
- Splitting

- Suppose each event in the Poisson process is divided as Type I, with probability p and Type II, with probability $1-p$ and $N_1(t)$ and $N_2(t)$ denote the random variables respectively
- Then, $N(t) = N_1(t) + N_2(t)$, where $N_1(t)$ and $N_2(t)$ are both Poisson processes with rates λp and $\lambda(1-p)$



- Pooling

- Similarly, suppose two Poisson processes are pooled together
- Then, $N_1(t) + N_2(t) = N(t)$, where $N(t)$ is a Poisson processes with rates $\lambda_1 + \lambda_2$



Poisson Distribution vs Exponential Distribution

- The Poisson distribution gives the probabilities for the number of events taking place in the given (fixed) time period
- The exponential distribution gives the probabilities for times between the events
- Both of these events occur randomly in time at a constant average rate, λ
- Examples:
 - Inter-arrival time and service time **are exponentially distributed** (not Poisson distributed)
 - The arrivals **follow a Poisson distribution** with λ arrivals in a time period.

Memoryless assumption

Poisson process and exponential times are related but not interchangeable

Poisson Distribution

Counts arrivals in a fixed interval
Parameter λ is the mean count per unit time
Mean and variance are both λ

Exponential Distribution

Models a time such as an inter-arrival or service time
Mean and standard deviation are $1/\mu$
Has the memoryless property

Practical Check

Use Poisson for number of arrivals
Use exponential for time between arrivals or service
Check assumptions before solving

Example: $\lambda = 4$ visitors/minute means 2 visitors in 30 seconds on average

Kendall Notation (1953)

British Statistician – David G. Kendall

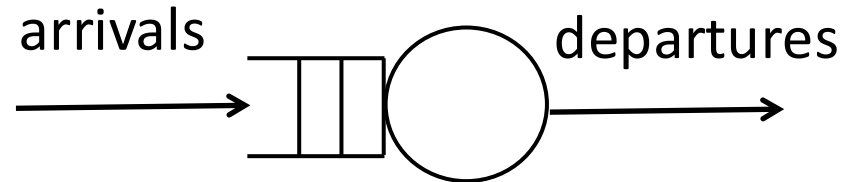
- describes characteristics of common queueing systems
- Kendall notations:

$A/S/m/K/N/SD$

where

- A is the inter-arrival time distribution
- S is the service time distribution
- m is the number of servers
- K is the system's capacity (number of places in the queue + number of servers)
- N is the number of customers in the source or population size
- SD is the service discipline

Classification of Single-Server Queues



		service time distribution	
		exponential	general
inter-arrival time distribution	exponential	M/M/1	M/G/1
	general	G/M/1	G/G/1

M: memoryless, exponential arrivals

G: general, or any probability distribution

Roadmap

- For simplicity,
 - Focus on a case of the Markov model, the birth-death model where state transitions are restricted to **neighboring states**
 - Single queue systems (with exponential inter-arrival time and exponential service time)
 - Single server
 - Multiple servers

Solving Markov Models

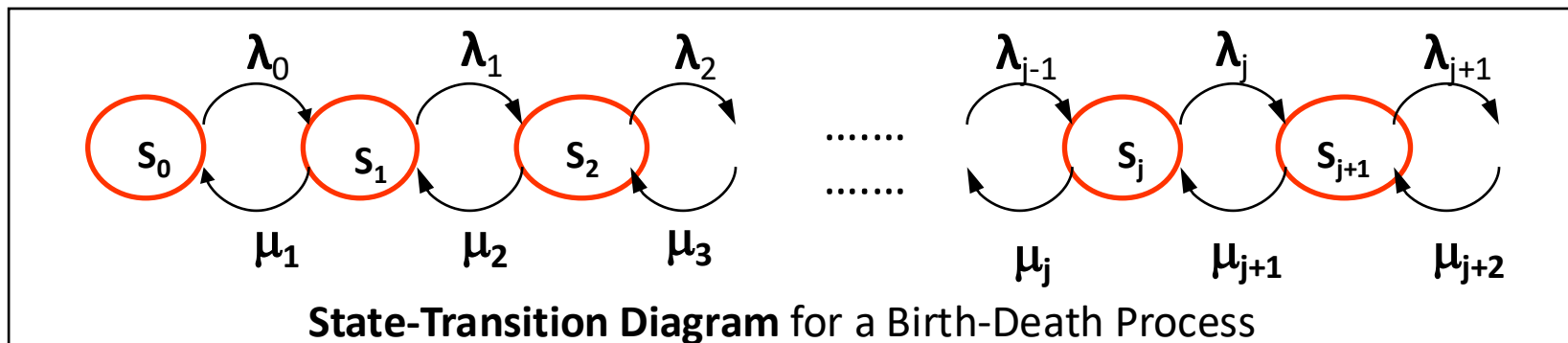
- Identify all possible system states to be modeled
- Determine transition rates between states
- Abstract a set of linear (balanced) equations and solve the steady-state probabilities of being in each state
- Use steady-state probabilities to find other metrics of interest – response time, utilization, ...

General Birth-Death Process

- Birth-death process is a special case of continuous-time Markov process where the state transitions are of only two types:
 - births, which increase the state variable by one; and
 - deaths, which decrease the state by one
- Used to study the evolution of bacteria, the number of customers in line at the supermarket, the number of packets queueing at a router, etc.

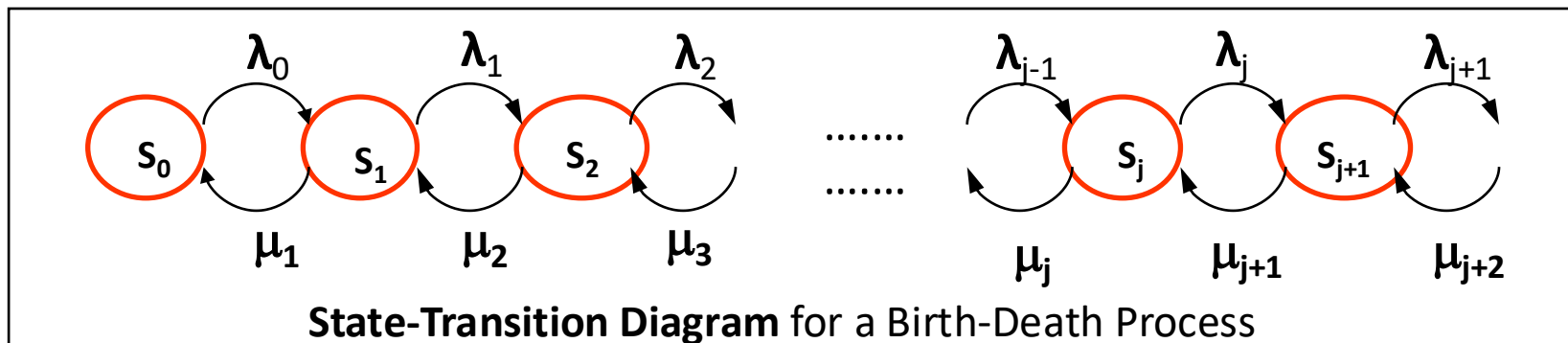
General Birth-Death Process Queueing Model

- Model a system where jobs arrive **one** at a time, where S is the number of jobs in the system
- Objective: determine P_n (probability of n jobs in the system) and P_0 (probability of zero jobs in the system) and subsequently other performance metrics



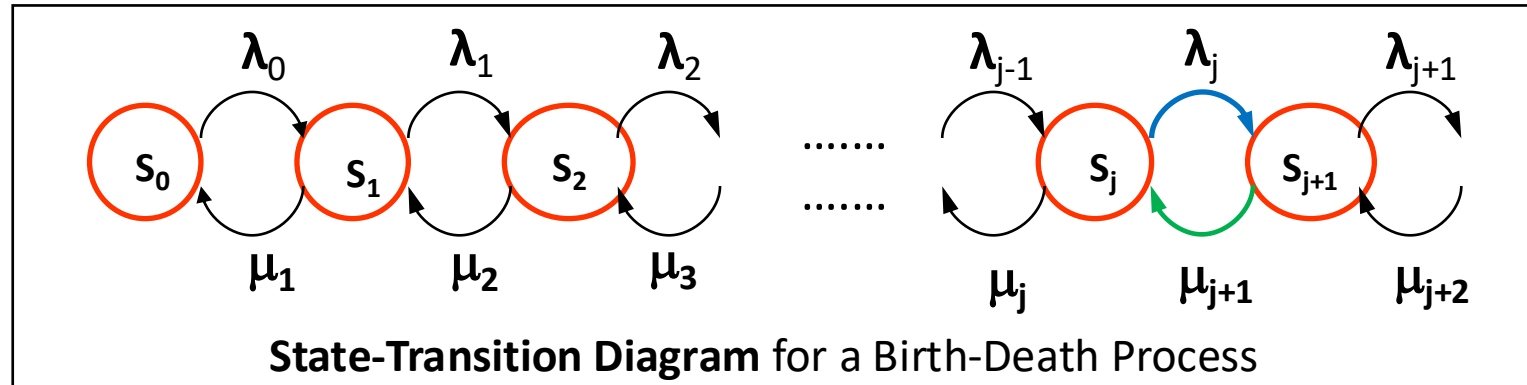
General Birth-Death Process Queueing Model

- Assumptions: exponential inter-arrival times and service times
- Suppose that whenever there are n people in the system, then
 - new arrivals enter the system at an exponential rate λ_n , and
 - people leave the system at an exponential rate μ_n
- That is, whenever there are i persons in the system, the time until the next arrival is exponentially distributed with mean $1/\lambda_i$ and is independent of the time until the next departure, which is itself exponentially distributed with mean $1/\mu_i$.



General Birth-Death Process Queueing Model

- A birth is a job arrival (change from state S_j to S_{j+1}), and a death occurs when a job leaves (departure) the system after completing (change from state S_j to S_{j-1}) service



When the system is in state S_n , it has n jobs in it.

- The new arrivals take place at a rate λ_n , and service rate is μ_n .
- Steady-state probability is $P_n = \frac{\lambda_0 \lambda_1 \dots \lambda_{n-1}}{\mu_1 \mu_2 \dots \mu_n} P_0$ $n = 1, 2, \dots, \infty$, where P_0 is the probability of being in state S_0

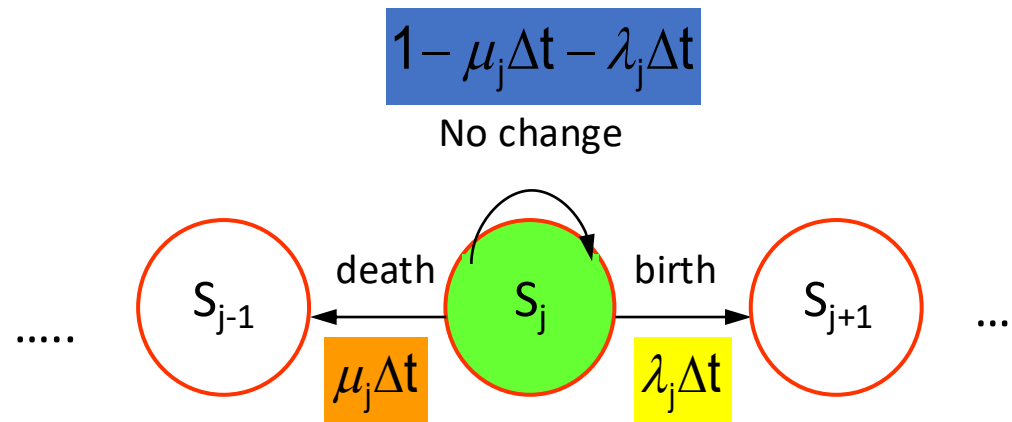
General Birth-Death Process Queueing Model

- Proof:

Suppose the system is in state S_j at time t

There are j jobs in the system

Consider a time interval $(t, t+\Delta t)$, where $\Delta t \rightarrow 0$



$$P_r[\text{one arrival} / \Delta t] = \lambda_j \Delta t$$

$$P_r[\text{one departure} / \Delta t] = \mu_j \Delta t$$

$$P_r[\text{no change} / \Delta t] = (1 - \lambda_j \Delta t)(1 - \mu_j \Delta t) \\ = 1 - \lambda_j \Delta t - \mu_j \Delta t$$

$$\text{when } \Delta t \rightarrow 0, (\Delta t)^2 \rightarrow 0$$

- Note: if Δt is small, the chance of two events occurring during this interval approaches zero

General Birth-Death Process Queueing Model

If $P_j(t)$ denotes the probability of being in state S_j at time t , then

A set of linear equations

$$\left\{ \begin{array}{l} P_0(t + \Delta t) = (1 - \lambda_0 \Delta t) P_0(t) + \mu_1 \Delta t P_1(t) \\ P_1(t + \Delta t) = \lambda_0 \Delta t P_0(t) + (1 - \mu_1 \Delta t - \lambda_1 \Delta t) P_1(t) + \mu_2 \Delta t P_2(t) \\ P_2(t + \Delta t) = \lambda_1 \Delta t P_1(t) + (1 - \mu_2 \Delta t - \lambda_2 \Delta t) P_2(t) + \mu_3 \Delta t P_3(t) \\ \vdots \\ P_j(t + \Delta t) = \lambda_{j-1} \Delta t P_{j-1}(t) + (1 - \mu_j \Delta t - \lambda_j \Delta t) P_j(t) + \mu_{j+1} \Delta t P_{j+1}(t) \end{array} \right.$$

- Next, solve equations to derive steady-state probabilities, P_n and P_0

Steady state vs Transient State

- Rule for all queues (lecture 3): **stability condition** - a system is stable when its long-run behaviour remains bounded
- The **steady state of a queueing system** is the state where the probability of the number of jobs in the system is independent of t
 - **FLOW RATE IN = FLOW RATE OUT** (for the system)
 - At steady state, long-run average arrivals and departures balance, and the relevant averages are finite (lecture 3)
- If $P_n(t)$ depends upon t , the queueing system is said to be in the **transient state**
- Note:
 - **Stability** prevents the queue from growing without bound
 - **Steady state** means the measured averages have settled into a long-run regime

General Birth-Death Process Queueing Model

- The j^{th} equation is as follows:

$$\lim_{\Delta t \rightarrow 0} \frac{P_j(t + \Delta t) - P_j(t)}{\Delta t} = \lambda_{j-1}P_{j-1}(t) - (\mu_j + \lambda_j)P_j(t) + \mu_{j+1}P_{j+1}(t)$$
$$\frac{dP_j(t)}{dt} = \lambda_{j-1}P_{j-1}(t) - (\mu_j + \lambda_j)P_j(t) + \mu_{j+1}P_{j+1}(t)$$

Assuming $P_j(t)$ is independent of time t

$$\therefore \lim_{t \rightarrow \infty} P_j(t) = P_j$$

At steady - state :

$$\lim_{t \rightarrow \infty} \frac{dP_j(t)}{dt} = \frac{dP_j}{dt} = 0$$

General Birth-Death Process Queueing Model

- Therefore, the j^{th} equation becomes:

$$0 = \lambda_{j-1}P_{j-1} - (\mu_j + \lambda_j)P_j + \mu_{j+1}P_{j+1}$$

$$P_{j+1} = \left(\frac{\mu_j + \lambda_j}{\mu_{j+1}} \right) P_j - \frac{\lambda_{j-1}}{\mu_{j+1}} P_{j-1}, j = 1, 2, \dots$$

$$P_1 = \frac{\lambda_0}{\mu_1} P_0$$

General Birth-Death Process Queueing Model

$$P_n = \frac{\lambda_0 \lambda_1 \dots \lambda_{n-1}}{\mu_1 \mu_2 \dots \mu_n} P_0 \quad n$$

- For state S_{n-1} :

$$= 1, 2, \dots, \infty$$

$$\text{Let } C_n = \frac{\lambda_0 \lambda_1 \dots \lambda_{n-1}}{\mu_1 \mu_2 \dots \mu_n}$$

$$P_n = C_n P_0$$

$$\text{Since } \sum_{n=0}^{\infty} P_n = 1$$

$$\therefore P_0 + \sum_{n=1}^{\infty} P_n = 1$$

$$P_0 + \sum_{n=1}^{\infty} C_n P_0 = 1$$

$$P_0 \left(1 + \sum_{n=1}^{\infty} C_n \right) = 1$$

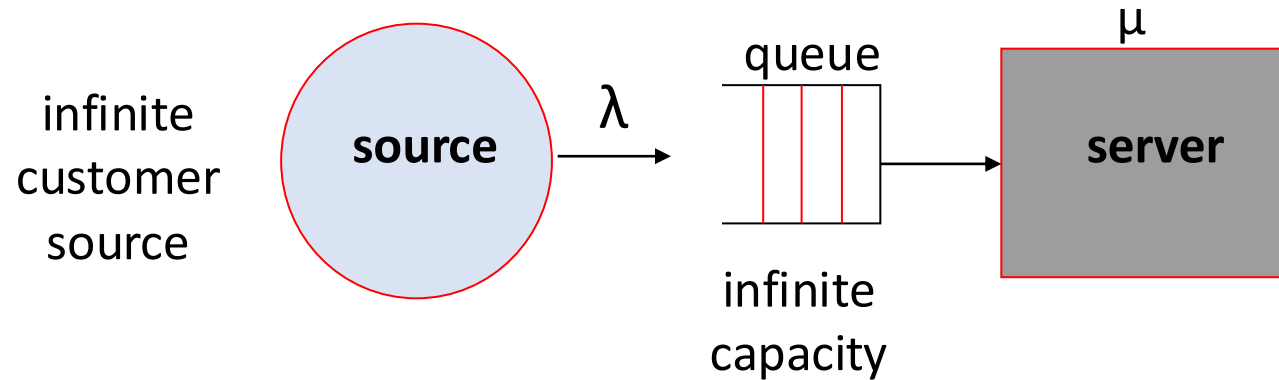
$$P_0 = \frac{1}{1 + \sum_{n=1}^{\infty} C_n}$$

Approach

Key steps

1. Identify all **possible states** to be modeled and draw a state-transition diagram
2. Form **balance equations**, i.e., a set of linear equations
3. Solve equations to derive **steady-state probabilities**, P_n and P_0
4. Use the steady-state probabilities to **derive other metrics** – response time, utilization, ...

M/M/1 Queueing System (M/M/1/ ∞ / ∞ /FCFS)



**Non-Preemptive Single-server
Open Queueing Model**

At steady-state:

FLOW RATE IN = FLOW RATE OUT

M/M/1 Queueing System

Balance Equation:

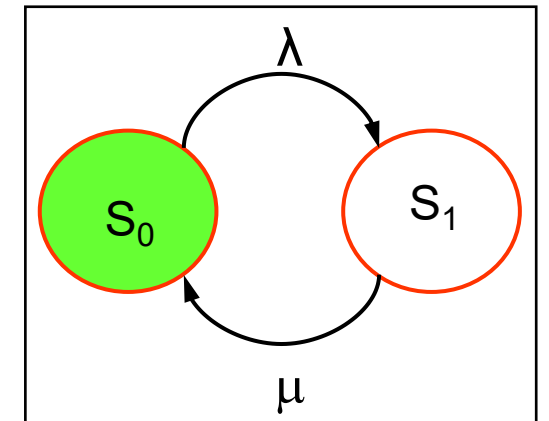
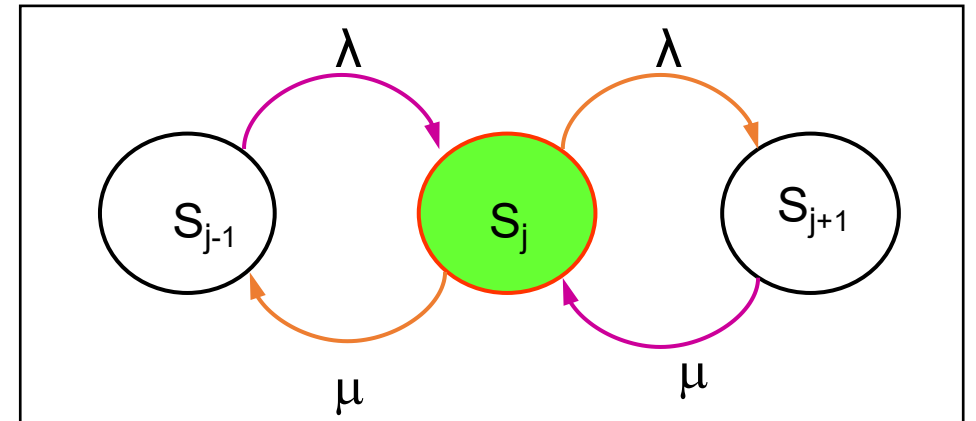
for $j \geq 1$:

$$\lambda P_{j-1} + \mu P_{j+1} = \lambda P_j + \mu P_j$$

for $j = 0$:

$$\begin{aligned} \lambda P_0 &= \mu P_1 \\ \Rightarrow P_1 &= \lambda / \mu P_0 \\ &= \rho P_0 \end{aligned}$$

where λ / μ is the traffic intensity, or utilization and is denoted by ρ



Extra reference: <https://youtu.be/rBIQmwaoZfs?si=du9MUEKExS0yr64j>

M/M/1 Queueing System

For $j = 1$

$$\begin{aligned}
 \lambda P_0 + \mu P_2 &= \lambda P_1 + \mu P_1 \\
 \mu P_2 &= (\lambda + \mu) P_1 - \lambda P_0 \\
 &= (\lambda + \mu) \rho P_0 - \lambda P_0 \\
 &= (\lambda^2 / \mu) P_0 + \lambda P_0 - \lambda P_0 \\
 \therefore P_2 &= \rho^2 P_0
 \end{aligned}$$

By induction:

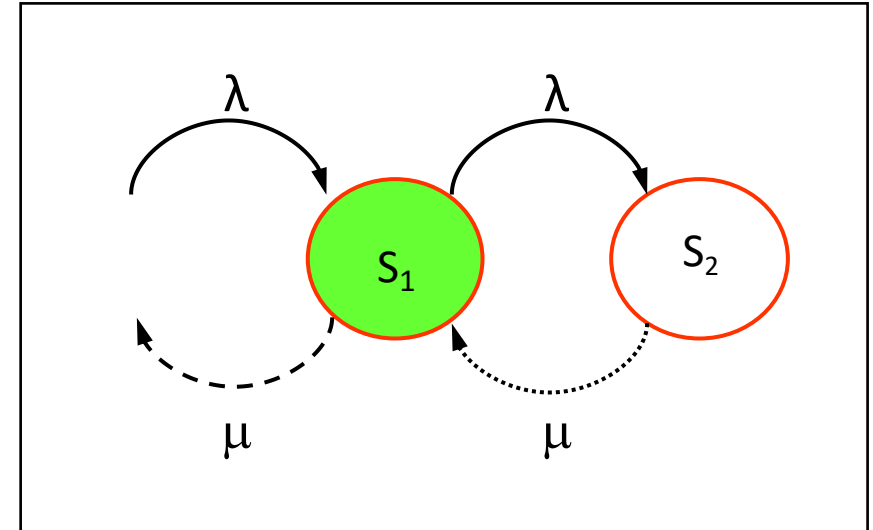
$$P_n = \rho^n P_0$$

$$\text{Since } \sum_{n=0}^{\infty} P_n = 1$$

$$\begin{aligned}
 \therefore P_0 &= \frac{1}{1 + \rho + \rho^2 + \dots + \rho^{\infty}} \\
 P_0 &= 1 - \rho \quad \text{and } P_n = (1 - \rho) \rho^n
 \end{aligned}$$

Utilization of the server:

$$U = 1 - P_0 \Rightarrow \mathbf{U = \rho}$$



Conservation of total probability!

$$P_0 + P_1 + P_2 + \dots = 1$$

$$P_0 + \rho P_0 + \rho^2 P_0 + \dots = 1$$

M/M/1 Queueing System

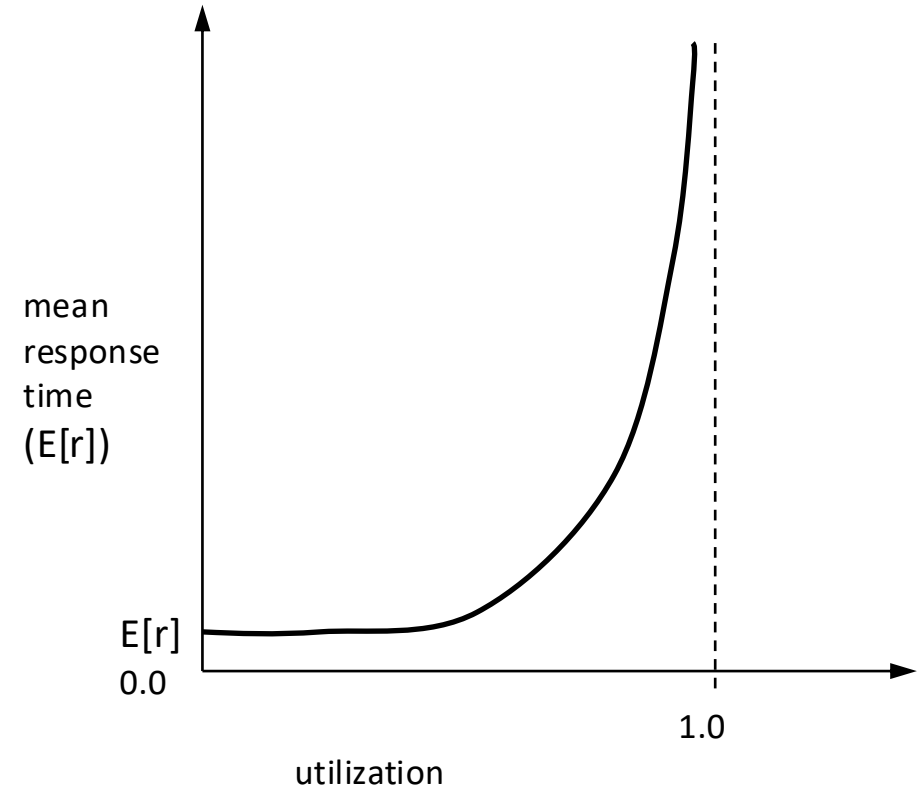
- Mean number of jobs in the system:

$$\begin{aligned} E[n] &= \sum_{n=1}^{\infty} n P_n \\ &= \sum_{n=1}^{\infty} n (1-\rho) \rho^n \end{aligned}$$

$$E[n] = \frac{\rho}{1-\rho}$$

Traffic Intensity ($\rho = \lambda / \mu$)

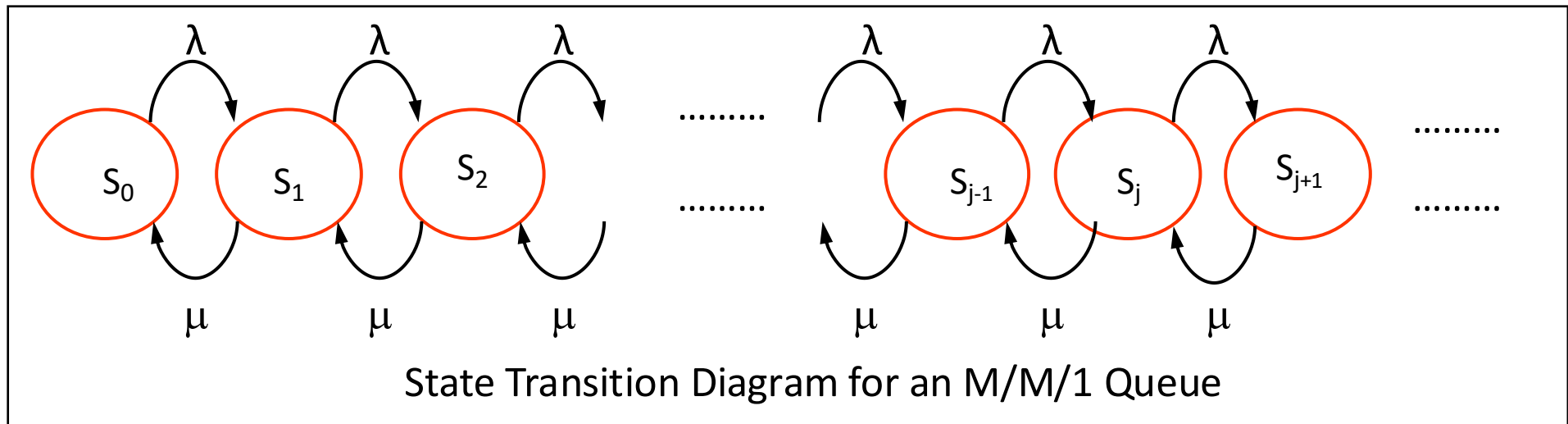
- Rate at which work enters the system
- Product of arrival rate and mean service time
- Determines minimum number of servers to keep up with the incoming stream of jobs
- Though dimensionless quantity, stated usually in units of Erlangs
- For a single server, lossless system (e.g., M/M/1), this is equal to server utilization ρ



Mean response time vs offered load
for M/M/1 queue

M/M/1 Queueing System

Alternatively, from birth-death processes:



$$\lambda_n = \lambda, \quad n = 0, 1, 2, \dots, \infty$$

$$\mu_n = \mu, \quad n = 1, 2, \dots, \infty$$

M/M/1 Queueing System

$$\text{Since } P_n = \frac{\lambda_0 \lambda_1 \dots \lambda_{n-1}}{\mu_1 \mu_2 \dots \mu_n} P_0, \quad n = 1, 2, \dots, \infty$$

$$P_n = \left(\frac{\lambda}{\mu}\right)^n P_0$$

$$\therefore P_n = \rho^n P_0 \quad [\text{traffic intensity : } \rho = \left(\frac{\lambda}{\mu}\right)]$$

$$\text{Since } P_n = \frac{1}{1 + \sum_{n=1}^{\infty} C_n}, \quad \text{where } C_n = \left(\frac{\lambda}{\mu}\right)^n = \rho^n$$

$$P_n = \frac{1}{1 + \rho + \rho^2 + \dots + \rho^{\infty}}$$

$$\therefore P_0 = 1 - \rho$$

$$P_0 = 1 - \rho$$

$$P_n = (1 - \rho)\rho^n$$

when $\rho < 1$

Example (1)

On a network gateway, measurements show that the **packets arrive at a mean rate of 125 packets per second (pps) and the gateway takes about 2 milliseconds** to forward them. Analyse the gateway using an M/M/1 model. What is the probability of buffer overflow if the gateway had only 12 buffers? How many buffers do we need to keep packet loss below one packet per million?

$$\lambda = 125 \text{ pps}$$

$$\mu = 1/0.002 = 500 \text{ pps}$$

$$\text{Utilization } \rho = \lambda / \mu = 0.25$$

Probability of buffer overflow

$$= P(13 \text{ or more packets in gateway})$$

$$= \rho^{13} = 1.49 \times 10^{-8}$$

≈ 15 packets per billion packets

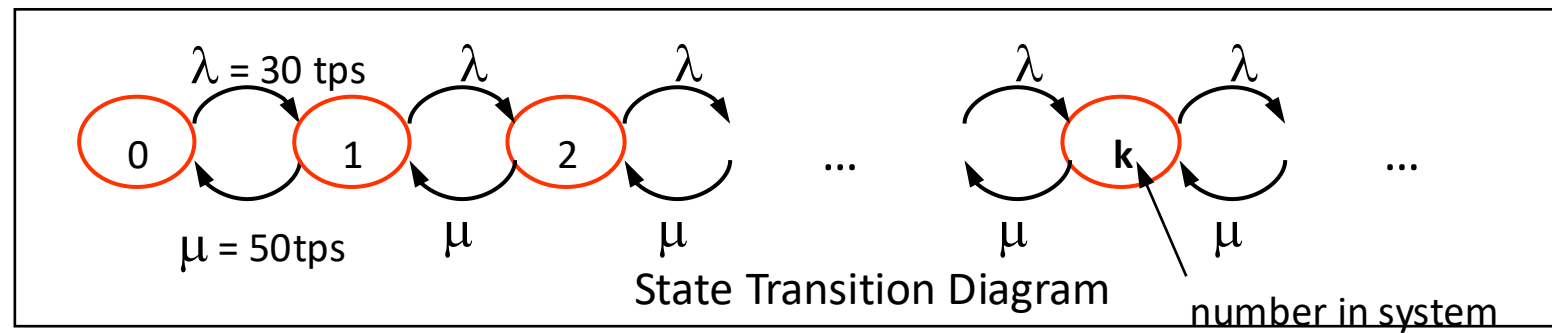
To limit the probability of packet loss to less than 10^{-6} ,

$$\rho^n \leq 10^{-6}$$

$$n > \log(10^{-6}) / \log(0.25) = 9.96$$

We need about **10 buffers**.

Example (2)



Suppose an average of **30 tps** arrive at a computer system. Each transaction requires an average of **20 msec** of processing from the central processor. What is the current performance of the system (e.g. processor utilization, system response time, system throughput, mean number of transactions in the system)?

Workload parameter: arrival rate of transaction (λ)

System parameter: service rate of the server (μ)

Given

$$\lambda = 30 \text{ tps}$$

$$\mu = 1 \text{ transaction} / 20 \text{ milliseconds} = 50 \text{ tps}$$

Applying Stochastic Analysis:

State description is the number of customers in the system.

$$P_0 = 1 - \lambda/\mu$$

$$= 40 \%$$

$$P_1 = (1 - \lambda/\mu) \lambda/\mu$$

$$= 24 \%$$

$$P_3 = 14\% \dots$$

$$P_0 + P_1 + P_2 + \dots = 1$$

Utilization $= P_1 + P_2 + \dots$
 $= 1 - P_0$
 $= 60 \%$

Throughput $= 0P_0 + \mu P_1 + \mu P_2 + \dots$
 $= \mu (P_1 + P_2 + \dots)$
 $= \mu \times \text{utilization}$
 $= \lambda = 30 \text{ tps}$

😊 throughput = arrival rate (flow rate in = flow rate out)

Mean no. of trans in sys, $E[n]$

$$= 0P_0 + 1P_1 + 2P_2 + \dots$$

$$= \sum_{k=1}^{\infty} kP_k = \sum_{k=1}^{\infty} k \left(\frac{\lambda}{\mu}\right)^k \left(1 - \frac{\lambda}{\mu}\right) \dots$$

$$= \frac{\lambda}{\mu - \lambda} = 1.5 \text{ transactions}$$

Response Time, $E[r]$ $= \left(\frac{E[n]}{\text{throughput}} \right)$

$$= \underbrace{\frac{1}{\mu} \times \frac{\lambda}{\mu - \lambda}}_{\text{Time to finish those in queue}} + \underbrace{\frac{1}{\mu}}_{\text{My service time}} = \frac{1}{\mu - \lambda}$$

$$= \frac{1}{20} \text{ second}$$

Summary:

utilization = 60 %

$E[n] = 1.5$ transactions

throughput = 30 tps

$E[r] = 1/20 \text{ sec}$

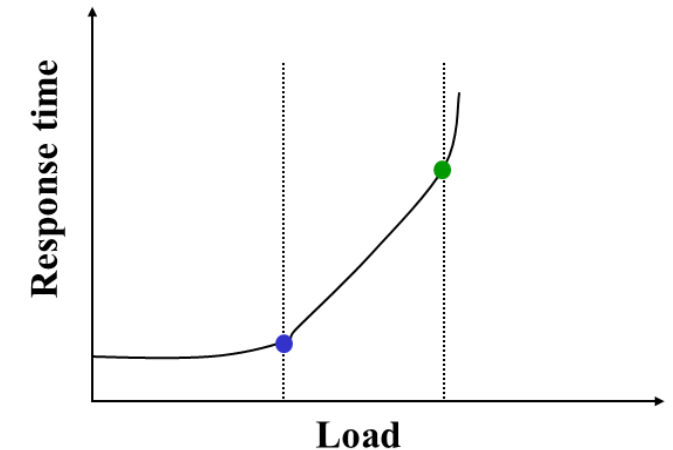
(i) What is the predicted performance if the workload increases by 50% more transactions per second ?

New transaction arrival rate, $\lambda' = 45$ tps

Assuming there are no secondary effect, and μ remains 50 tps

Recompute the measures:

utilization	= 90%	50% ↑
throughput	= 45 tps	50% ↑
$E[n]$	= 9 transactions	500%
$E[r]$	= 1/5 sec	300%



$E[n]$ and $E[r]$ are more sensitive
to changes in the system parameters !

(ii) What is the predicted performance if the workload increases by 50 % and the processor is upgraded to run 50% faster ?

As before, $\lambda' = 45$ tps

New processor can handle 50% more transactions:

$$\mu' = 75 \text{ tps}$$

Recompute the measures:

utilization	= 60%	unchanged
throughput	= 45 tps	50% ↑
$E[n]$	= 1.5 transactions	unchanged
response time	= 1/30 sec	33% ↓

Implications:

Processor speed is a dominating factor and increasing processor speed by 50% more than offset a 50% increase in the workload

Thus, to maintain a given response time level as the workload increases by p%, a less than p% faster processor is required

M/M/1

- Parameters:

λ = arrival rate in jobs per unit time

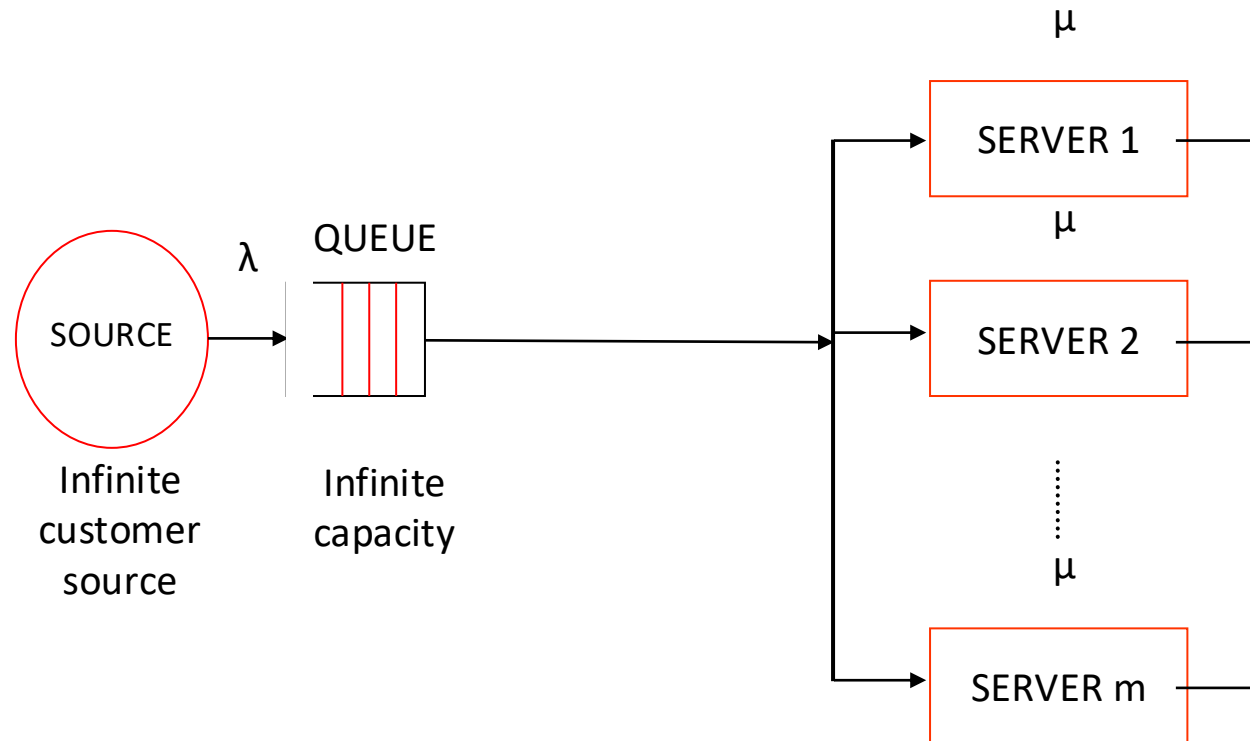
μ = service rate in jobs per unit time

- Traffic intensity: $\rho = \frac{\lambda}{\mu}$
- Stability condition: $\rho < 1$
- Probability of zero jobs in the system: $p_0 = 1 - \rho$
- Probability of n jobs in the system: $p_n = (1 - \rho)\rho^n$
- Utilization: $U = 1 - p_0 = \rho$
- Mean number of jobs in the system: $E[n] = \frac{\rho}{1-\rho}$
- Variance of number of jobs in the system: $Var[n] = \frac{\rho}{(1-\rho)^2}$
- Probability of k jobs in the queue:

$$P(n_q = k) = \begin{cases} 1 - \rho^2, & k = 0 \\ (1 - \rho)\rho^{k+1}, & k > 0 \end{cases}$$
- Mean number of jobs in the queue: $E[n_q] = \frac{\rho^2}{(1-\rho)}$
- Variance of number of jobs in the queue: $Var[n_q] = \frac{\rho^2(1+\rho-\rho^2)}{(1-\rho)^2}$

- Cumulative distribution function of the response time: $F[r] = 1 - e^{-r\mu(1-\rho)}$
- Mean response time: $E[r] = \frac{1}{\mu(1-\rho)}$
- Variance of the response time: $Var[r] = \frac{1}{\mu^2(1-\rho)^2}$
- q-Percentile of the response time: $E[r] \ln\left(\frac{100}{100-q}\right)$
- 90-Percentile of the response time: $2.3E[r]$
- Cumulative distribution function of waiting time: $F[w] = 1 - \rho e^{-\mu w(1-\rho)}$
- Mean waiting time: $E[w] = \frac{\rho}{\mu(1-\rho)}$
- Variance of the waiting time: $Var[w] = \frac{(2-\rho)\rho}{[\mu^2(1-\rho)^2]}$
- q-percentile of the waiting time: $\max(0, \frac{E[w]}{\rho} \ln[\frac{100\rho}{100-q}])$
- 90-percentile of the waiting time: $\max(0, \frac{E[w]}{\rho} \ln[10\rho])$
- Probability of finding n or more jobs in the system: ρ^n
- Probability of serving n jobs in one busy period: $\frac{1}{n} \binom{2n-2}{n-1} \frac{\rho^{n-1}}{(1+\rho)^{2n-1}}$
- Mean number of jobs served in one busy period: $\frac{1}{1-\rho}$
- Variance of number of jobs served in one busy period: $\frac{\rho(1+\rho)}{(1-\rho)^3}$
- Mean busy period duration: $\frac{1}{\mu(1-\rho)}$
- Variance of the busy period: $\frac{1}{\mu^2(1-\rho)^3} - \frac{1}{\mu^2(1-\rho)^2}$

M/M/m Queueing System (M/M/ m / ∞ / ∞ /FCFS)



Non-Preemptive Multiple-server
Open Queueing Model

M/M/**m** Queueing System

From the general birth-death process:

$$\lambda_n = \lambda \quad n = 0, 1, 2, \dots, \infty$$

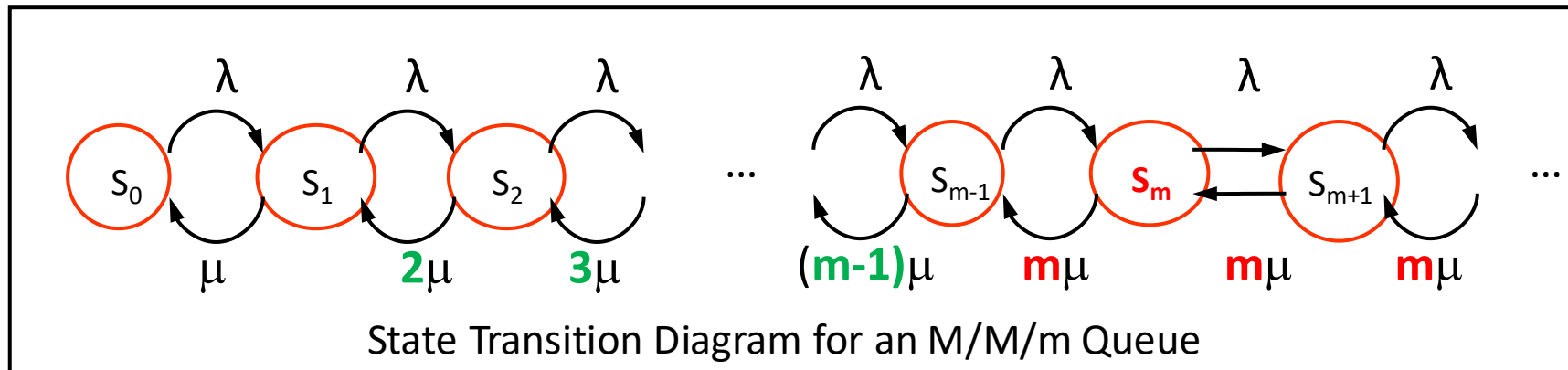
$$\mu_n = \begin{cases} n\mu & n = 1, 2, \dots, m-1 \\ m\mu & n = m, m+1, \dots, \infty \end{cases}$$

$$P_n = \begin{cases} \frac{(m\rho)^n}{n!} P_0, & n = 1, 2, \dots, m-1 \\ \frac{\rho^n m^m}{m!} P_0, & n = m, m+1, \dots, \infty \end{cases}$$

$$P_0 = \left[1 + \frac{(m\rho)^m}{m!(1-\rho)} + \sum_{n=1}^{m-1} \frac{(m\rho)^n}{n!} \right]^{-1} \quad \rho = \lambda / m\mu$$

[Exponential distribution property:

When the mean service rate per server is μ , then the overall mean service rate for m busy servers is $m\mu$.]



M/M/m

- Parameters:

λ = arrival rate in jobs per unit time

μ = service rate in jobs per unit time

m = number of servers

- Traffic intensity: $\rho = \frac{\lambda}{m\mu}$
- Stability condition: $\rho < 1$
- Probability of zero jobs in the system: $p_0 = \left[1 + \frac{(m\rho)^m}{m!(1-\rho)} + \sum_{n=1}^{m-1} \frac{(m\rho)^n}{n!} \right]^{-1}$
- Probability of n jobs in the system: $p_n = \begin{cases} \frac{p_0(m\rho)^n}{n!}, & n < m \\ \frac{p_0\rho^n m^m}{m!}, & n \geq m \end{cases}$
- Utilization: $U = \frac{\lambda}{m\mu} = \rho$
- Probability of queueing: $\varrho = P(\geq m \text{ jobs}) = \frac{(m\rho)^m}{m!(1-\rho)} p_0$
- Mean number of jobs in the system: $E[n] = m\rho + \frac{\rho\varrho}{1-\rho}$
- Variance of number of jobs in the system: $Var[n] = m\rho + \rho\varrho \left[\frac{1+\rho-\rho\varrho}{(1-\rho)^2} + m \right]$
- Mean number of jobs in the queue: $E[n_q] = \frac{\rho\varrho}{(1-\rho)}$
- Variance of number of jobs in the queue: $Var[n_q] = \frac{\rho\varrho(1+\rho-\rho\varrho)}{(1-\rho)^2}$
- Mean response time: $E[r] = \frac{1}{\mu} \left(1 + \frac{\varrho}{m(1-\rho)} \right)$
- Variance of the response time: $Var[r] = \frac{1}{\mu^2} \left[1 + \frac{\varrho(1-\varrho)}{m^2(1-\rho)^2} \right]$
- Cumulative distribution function of waiting time: $F[w] = 1 - \varrho e^{-m\mu w(1-\rho)}$
- Mean waiting time: $E[w] = \frac{E[n_q]}{\lambda} = \frac{\varrho}{m\mu(1-\rho)}$
- Variance of the waiting time: $Var[w] = \frac{\varrho(2-\rho)}{[m^2\mu^2(1-\rho)^2]}$
- q -percentile of the waiting time: $\max(0, \frac{E[w]}{\varrho} \ln[\frac{100\varrho}{100-q}])$
- 90-percentile of the waiting time: $\max(0, \frac{E[w]}{\varrho} \ln[10\varrho])$

Example

Students arrive at the university computer center in a Poisson manner at an average rate of 10 per hour. Each student spends an average of 20 minutes at the terminal, and the time can be assumed to be exponentially distributed. The center currently has five terminals. Some students have been complaining that waiting times are too long. Analyze the center using a queueing model.

The computer center can be modeled as an M/M/5 queueing system.

$$\lambda = 1/6 \text{ per minute}$$

$$\mu = 1/20 \text{ per minute}$$

$$\text{Traffic intensity } \rho = \lambda / m\mu$$

$$\rho = 0.67$$

Probability of queueing

(or all terminals being busy)

average terminal utilization

$$Q = \frac{(m\rho)^m}{m!(1-\rho)} P_0 = \frac{(5 * 0.67)^5}{5!(1-0.67)} 0.0318$$
$$= 0.33$$

average no. of students in the center:

$$\begin{aligned} E[n] &= m\rho + \rho q / (1 - \rho) \\ &= 4.0 \end{aligned}$$

average no. of students waiting in the queue:

$$\begin{aligned} E[n_q] &= \rho q / (1 - \rho) \\ &= 0.65 \end{aligned}$$

average no. of students using the terminals:

$$\begin{aligned} E[n_s] &= E[n] - E[n_q] \\ &= 4.0 - 0.65 \\ &= 3.35 \end{aligned}$$

mean waiting time:

$$\begin{aligned} E[w] &= q / [m\mu(1 - \rho)] \\ &= 0.33 / [5 \times 0.5 \times (1 - 0.67)] = 4 \text{ minutes} \end{aligned}$$

90-percentile of the waiting time is

$$\max \{ 0, E[w] / q \ln(10 q) \} = 14 \text{ minutes}$$

Thus 10 % of the students have to wait more than 14 minutes.

Example

The students would like to limit their **waiting time to an average of 2 minutes and no more than 5 minutes in 90% of the cases. Is it feasible?** If yes, then how many terminals are required?

$$\lambda = 0.167 \quad \mu = 0.05$$

analyze the system with $m = 6, 7, \dots$

With $m = 6$ terminals

$$\text{traffic intensity } \rho = 0.167 / (6 \times 0.05) = 0.556$$

$$\text{probability of all terminals being idle} = P_0 = 0.0346$$

$$\text{probability of all terminals being busy} = 0.15$$

$$\text{average waiting time} = \mathbf{1.1 \text{ minutes}}$$

$$\text{90-percentile of waiting time} = \mathbf{3.0}$$

Thus, with just one more terminal we will be able to satisfy the student's demand.

Example

Consider what would have happened if the five terminals were located in five different locations on the campus, thereby needing a separate queue for each.

(i) The system is modeled as five separate M/M/1 queues.

Arrival rate per terminal is $\lambda = 0.167/5 = 0.0333$, $m = 1$

$\mu = 0.05$

Traffic intensity $\rho = 0.0333/0.05 = 0.67$

Mean time spent in the terminal is $E[r] = 1/\mu / (1 - \rho) = 60$ minutes

$\text{Var}[r] = 1/\mu^2 / (1 - \rho)^2 = 3600$

(ii) M/M/5, Single Queue

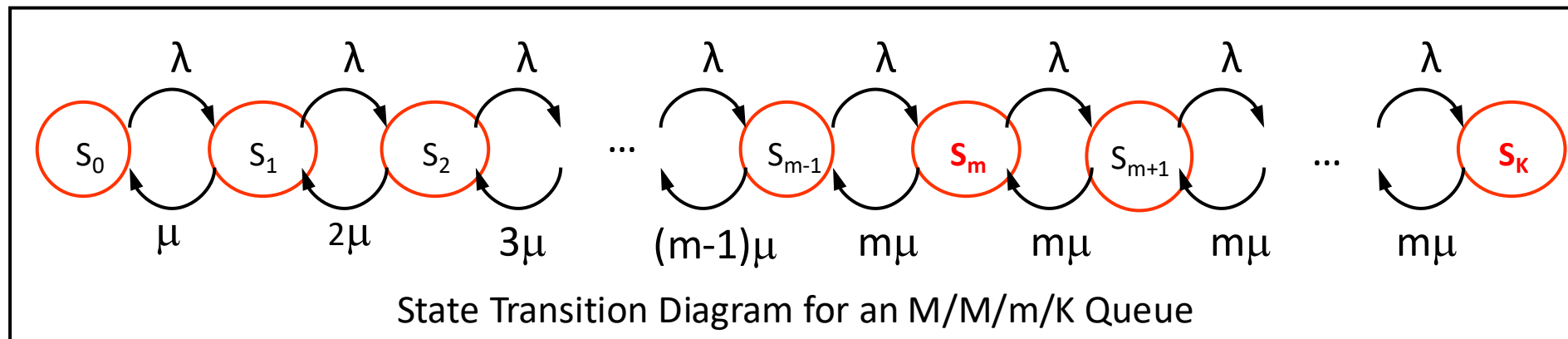
$E[r] = 24$ minutes

$\text{Var}[r] = 479$

if all jobs are identical, it is better to have just one queue than to have multiple queues
bank (tellers) vs supermarket checkouts

M/M/m/K Queueing System

- M/M/m/K is similar to the M/M/m queue except that the **number of customers in the system K is finite**.
- After **K** buffers are full, all arrivals are lost.
- **K** $\geq m$



$$\lambda_n = \lambda, \quad n = 0, 1, 2, \dots, K-1$$

$$\mu_n = \begin{cases} n\mu & n = 1, 2, \dots, m-1 \\ m\mu & n = m, m+1, \dots, K \end{cases}$$

$$P_n = \begin{cases} \frac{(m\rho)^n}{n!} P_0, & n = 1, 2, \dots, m-1 \\ \frac{m^m \rho^n}{m!} P_0, & n = m, m+1, \dots, K \end{cases}$$

$$P_0 = \left[1 + \frac{(1 - \rho^{B-m+1})(m\rho)^m}{m! (1 - \rho)} + \sum_{n=1}^{m-1} \frac{(m\rho)^n}{n!} \right]^{-1}$$

$$E[n] = \sum_{n=1}^K n P_n \quad \text{- mean no. of jobs in the system}$$

$$E[n_q] = \sum_{n=m+1}^K (n - m) P_n \quad \text{- mean no. of jobs in the queue}$$

Effective arrival rate (λ'):

$$\lambda' = \sum_{n=0}^{K-1} \lambda P_n = \lambda(1 - P_K)$$

when the system state $n=K$, all arrivals are lost.

$\lambda - \lambda' = \lambda P_K$ is the packet loss rate.

M/M/m/K

- Parameters:

λ = arrival rate in jobs per unit time

μ = service rate in jobs per unit time

m = number of servers

K = number of buffers (B), $K \geq m$

- Traffic intensity: $\rho = \frac{\lambda}{m\mu}$

- The system is always stable.

- Probability of zero jobs in the system:

$$p_0 = \left[1 + \frac{(1 - \rho^{K-m+1})(m\rho)^m}{m!(1 - \rho)} + \sum_{n=1}^{m-1} \frac{(m\rho)^n}{n!} \right]^{-1}$$

- For $m=1$: $p_0 = \begin{cases} \frac{1-\rho}{1-\rho^{K+1}}, \rho \neq 1 \\ \frac{1}{K+1}, \rho = 1 \end{cases}$

- Probability of n jobs in the system: $p_n = \begin{cases} \frac{p_0(m\rho)^n}{n!}, 0 \leq n < m \\ \frac{p_0\rho^n m^m}{m!}, m \leq n \leq K \end{cases}$

- Effective arrival rate in the system: $\lambda' = \sum_{n=0}^{K-1} \lambda p_n = \lambda(1 - p_K)$

- Utilization: $U = \frac{\lambda'}{m\mu} = \rho(1 - p_K)$

- For an M/M/m/m queue, the probability of a full system is: $p_m = \frac{(m\rho)^m}{m! \sum_{j=0}^m \frac{(m\rho)^j}{j!}} p_0$

- The loss rate is given by λp_K jobs per unit time.

- Mean number of jobs in the system: $E[n] = \sum_{n=1}^K n p_n$

- For $m=1$: $E[n_q] = \frac{\rho}{(1-\rho)} - \frac{\rho(1+K\rho^K)}{1-\rho^{K+1}}$

- Mean number of jobs in the queue: $E[n_q] = \sum_{n=m+1}^K (n - m) p_n$

- Mean response time: $E[r] = \frac{E[n]}{\lambda'} = \frac{E[n]}{\lambda(1-p_K)}$

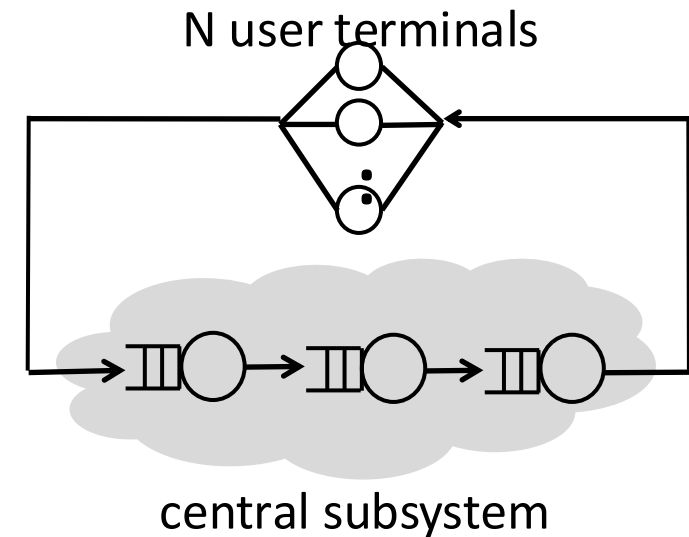
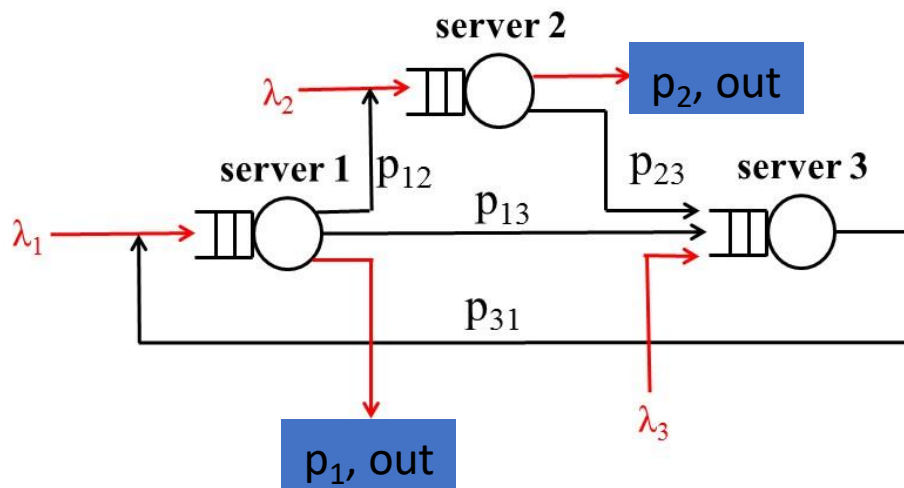
- Mean waiting time: $E[w] = \frac{E[n_q]}{\lambda(1-p_K)} = E[r] - \frac{1}{\mu}$

Product Form Queueing Networks

queueing networks

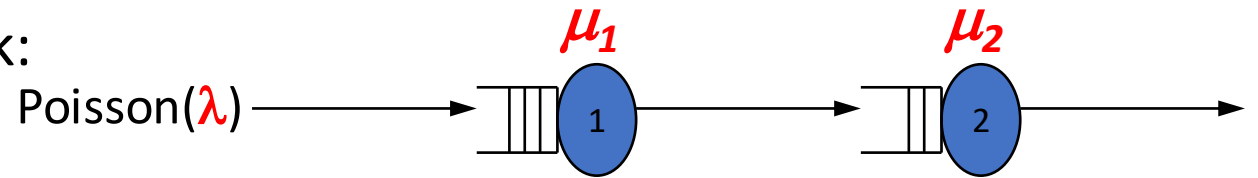
open
finite buffer,
infinite buffer

closed
interactive, batch



Product Form Queueing Networks

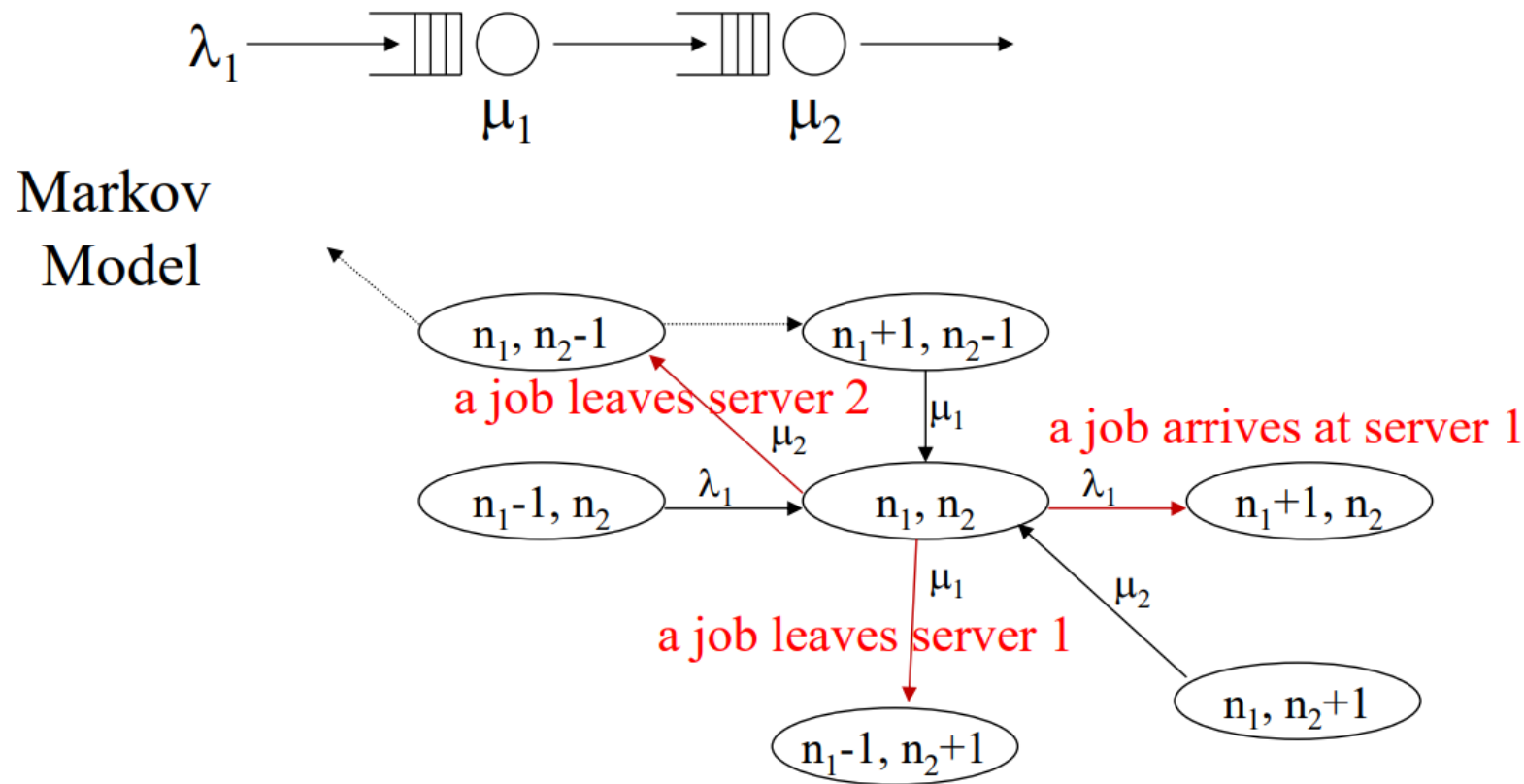
- Consider a two-stage tandem network:



Two-stage Tandem Network

- Poisson arrival process with arrival rate λ
- Service time distributions at both centers are **exponential** with service rates of μ_1 and μ_2
- The system can be modeled as a 2-D stochastic process:
 - States are specified by pairs (N_1, N_2) , $N_1 \geq 0$, $N_2 \geq 0$, where $N_i (i=1, 2)$ is the number of jobs at queue i at steady-state
 - State changes occur upon a completion of service at one of the two queues, or upon an external arrival
 - All inter-event times are exponential $\Rightarrow$ stochastic process is a 2-D CTMC

State diagram for tandem queueing network with 2 servers



<https://people.cs.vt.edu/~irchen/5214/pdf/p129-149.pdf>

Product Form Queueing Networks

- Draw 2-D state transition diagram, derive and solve balanced equations to obtain steady-state probabilities:

$$p(n_1, n_2) = P[N_1=n_1, N_2=n_2] = (1 - \rho_1)\rho_1^{n_1} (1 - \rho_2)\rho_2^{n_2}$$

Where $\rho_i = \lambda / \mu_i$ ($i = 1, 2$)

- Condition for system stability is that both ρ_1 and ρ_2 are less than 1
- Equation (1) is called **a product form solution**
 - Note that **$N_1(t)$ and $N_2(t)$ are not independent during transient state**
 - However, at steady state, **$N_1(t)$ and $N_2(t)$ are independent**
 - $P[N_1=n_1, N_2=n_2] = P[N_1=n_1] \times P[N_2=n_2]$
and $P[N_i=n_i] = (1 - \rho_i)\rho_i^{n_i}$, $i = 1, 2$
 - **At steady-state, the two queues act as independent M/M/1 queues**
 - Thus, product form solution can be generalized to an m-stage tandem network

Product Form Queueing Networks

- A queueing network is a product form network, if the steady-state number of jobs in the system is:

$$p(n_1, n_2, \dots, n_M) = \frac{1}{G(N)} \prod_{i=1}^M f_i(n_i)$$

Where $f_i(n_i)$ is some function of the number of jobs at the i th queue
 $G(N)$ is a normalizing constant and a function of total number of jobs in the system

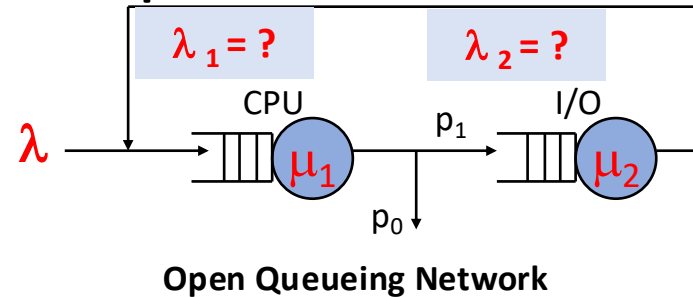
- Several techniques to solve product form queueing networks; however, non-product form networks are intractable

Examples of Product Form Queueing Networks

In increasing complexity:

- **Jackson networks**
 - Arbitrary open network of m -server, FCFS queues; exponentially distributed service times; independent Poisson external arrivals; Markovian routing (i.e., sequence of stations visited by each job is an irreducible DTMC)
- **Gordon–Newell networks**
 - Arbitrary closed networks of m -server FCFS queues; exponentially distributed service times; Markovian routing
- **BCMP networks**
 - See the classical paper - Baskett et al., “Open, Closed, and Mixed Networks of Queues with Different Classes of Customers”, Journal of ACM, 22(2), 248-260, 1975

Example: Open Jackson Network



- Two single-server queues
- Service times at CPU $\sim \exp(\mu_1)$ & I/O $\sim \exp(\mu_2)$
- Poisson arrival at CPU from outside
- Job from CPU goes to I/O with probability p_1 and exits the system with probability $p_0 = 1 - p_1$
- Job from I/O returns to CPU
- FCFS service at both CPU and I/O
- Jackson's result - the two queues behave like independent M/M/1 queues

- Hence: $p(n_1, n_2) = (1 - \rho_1) \rho_1^{n_1} (1 - \rho_2) \rho_2^{n_2}$
where $\rho_i = \lambda_i / \mu_i$, $i = 1, 2$

- We need to compute net arrival rates λ_1 & λ_2 into the two queues
 - At steady-state, departure rates from the two queues will also be λ_1 & λ_2 , respectively
 - Thus, net arrival rate at CPU: $= \lambda_1 = \lambda + \lambda_2$
 - Net arrival rate at I/O: $= \lambda_2 = \lambda_1 p_1$
 - Solving these two, we have

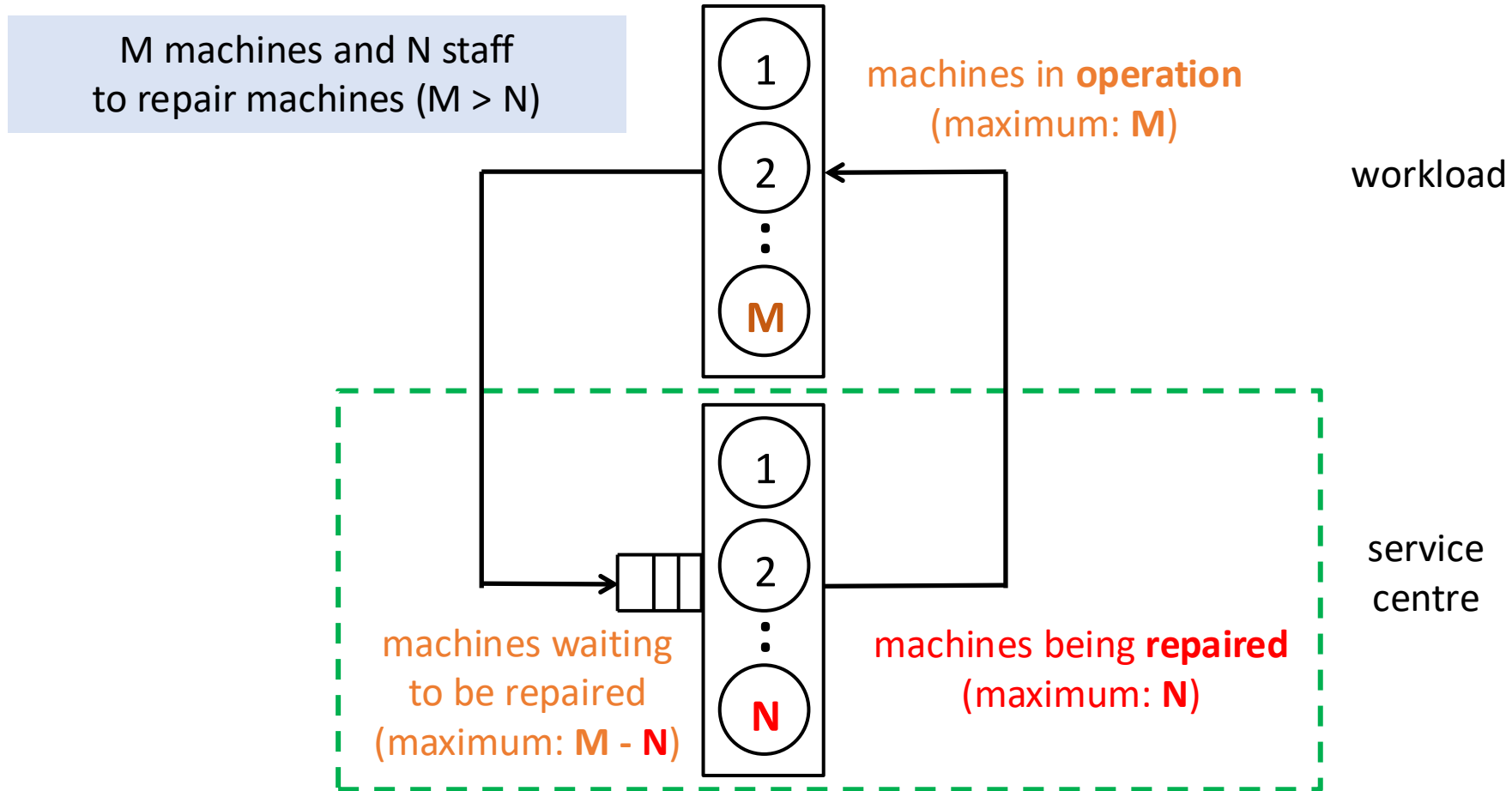
$$\lambda_1 = \lambda / p_0 \quad \text{and} \quad \lambda_2 = p_1 \lambda / p_0$$

Product Form Networks (Summary)

- Restrict to **exponential service times**, FCFS service discipline, and Poisson external arrivals (if any)
- For **non-exponential service times** and for service disciplines other than FCFS (solve using BCMP networks)
- Problem: obtain joint probability distribution of steady-state number of jobs in the m service centers:
 - **Open network** - easy to apply Jackson's result: product of the marginal distributions at individual service centers with appropriate arrival & service rates
 - **Closed network** complicated to obtain the normalizing constant $G(N)$ in the product form solution!
 - A technique called convolution algorithm can be used for computing $G(N)$ and various performance metrics for closed queueing networks

Example: Reliability of a Data Center

Reliability of a Data Center



Problem

- Automatic diagnostic system – “ping” to detect failure and to inform staff when failure is detected
- Machine is in one of two states: **operational** or **failed**
- **Failed machine**: waiting to be repaired by one of N repair staff or in the process of being repaired
- If time-to-next-failure and the repair time **are exponentially distributed**, we can solve this queueing problem using Markov chain!

Performance Questions

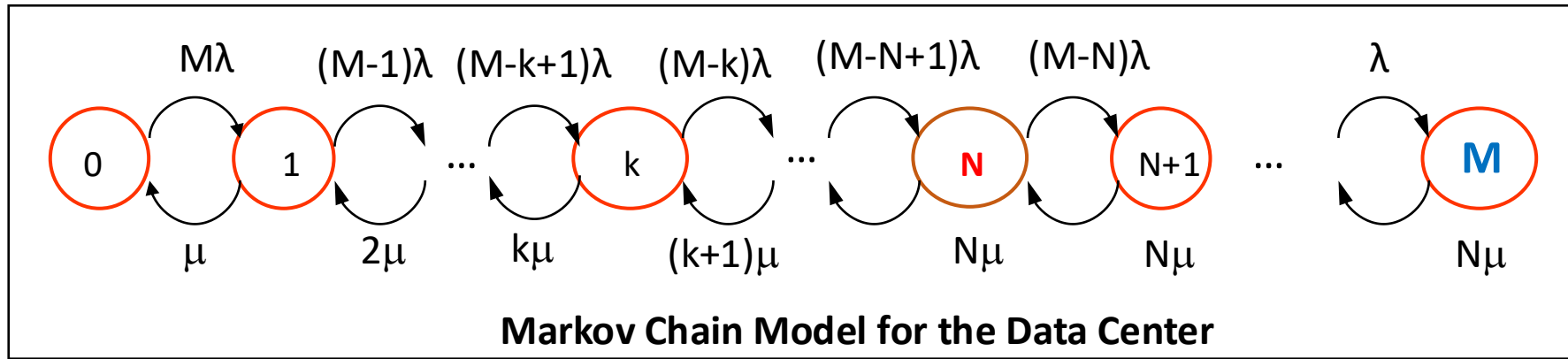
- What is the probability that exactly j ($j = 1, \dots, M$) machines are operational at any given time?
- What is the probability that at least j ($j = 1, \dots, M$) machines are operational at any given time?
- How many repair staff is required to guarantee that at least j ($j = 1, \dots, M$) machines are operational with a given probability?
- What is the effect of the size of the repair team on the mean-time-to-repair (MTTR) a machine?
What is the effect of the size of the repair team on the percentage of machine that can be expected to be operational at any given time?
- What is the effect of the average time it takes a member of the repair team to fix a machine (i.e., their skill level) on the overall MTTR (i.e., includes the time waiting to repair)? How does a repair person's skill level affect the percentage of operational machine?

Model Construction

Building a Model

- Assumptions: machines fail independently and at the same rate, all repair staff have the same skill level and thus the same repair rate
 - Let λ denotes the machine failure rate. Thus, $1/\lambda$ is the mean time to failure
 - Let μ denotes the repair rate
- States: let state k represent the number of failed machines (failed machines need to be repaired)
- State transitions:
 - transition from state k to $k + 1$ happens when a machine fails
 - transition from state k to $k - 1$ occurs when a machine is repaired
 - at state k , there are $M - N$ machines in operation and each has a failure rate of λ

Model Construction



- Aggregate failure rate, λ_k , at state k is

$$\lambda_k = (M - k)\lambda \quad k = 0, \dots, M - 1$$
- Aggregate repair rate, μ_k , at state k depends on whether all N repair staff are busy,

$$\mu_k = \begin{cases} k\mu & k = 1, \dots, N \\ N\mu & k = (N + 1), \dots, M \end{cases}$$

Model Construction

Solve the model (Markov Chain) for p_k , steady-state probability of being in state k ($k=0,\dots,M$), i.e., k machines failed

$$p_k = \begin{cases} p_0 \left(\frac{\lambda}{\mu}\right)^k \binom{M}{k} & k = 1, \dots, N \\ p_0 \left(\frac{\lambda}{\mu}\right)^k \binom{M}{k} \frac{N^{N-k} k!}{N!} & k = (N + 1), \dots, M \end{cases}$$

where p_0 is obtained by $\sum_{k=0}^M p_k = 1$. Hence,

$$p_0 = \left[\sum_{k=0}^N \left(\frac{\lambda}{\mu}\right)^k \binom{M}{k} + \sum_{k=N+1}^M \left(\frac{\lambda}{\mu}\right)^k \binom{M}{k} \frac{N^{N-k} k!}{N!} \right]^{-1}$$

Model Construction

- Average aggregate rate at which machines fail (mean failure rate):

$$\bar{X}_f = \sum_{k=0}^{M-1} \lambda_k \times p_k = \sum_{k=0}^{M-1} (M - k) \lambda p_k$$

- Mean time to repair:

$$\text{MTTR} = \frac{M}{\bar{X}_f} - \text{MTTF} = \frac{M}{\bar{X}_f} - 1/\lambda$$

- Average number of failed machines:

$$N_f = \bar{X}_f \times \text{MTTR} = M - \bar{X}_f \times \text{MTTF} = M - \bar{X}_f / \lambda$$

- Average number of operational machines:

$$N_o = M - N_f = \bar{X}_f \times \text{MTTF} = \bar{X}_f / \lambda$$

Model Interpretation

- Assume there are 120 machines in the datacenter with MTTF of 500 minutes ($\lambda = \frac{1}{500} = 0.002$ failures per minute) with average time of 20 minutes ($\mu = \frac{1}{20} = 0.05$ repairs per minute) to repair a machine
- Then we can answer the performance questions:
 1. What is the probability that exactly j ($j = 1. \dots, M$) machines are operational at any given time?
 2. What is the probability that at least j ($j = 1. \dots, M$) machines are operational at any given time?
 3. How many repair people are necessary to guarantee that at least j ($j = 1. \dots, M$) machines are operational with a given probability?
 4.

Model Interpretation

1. What is the probability that exactly j ($j = 1, \dots, M$) machines are operational at any given time?

The probability that j machines are operational is the probability that $M - j$ machines failed

$$p_{M-j} = \begin{cases} p_0 \left(\frac{\lambda}{\mu} \right)^{M-j} \binom{M}{M-j} & j = M - N, \dots, M - 1 \\ p_0 \left(\frac{\lambda}{\mu} \right)^{M-j} \binom{M}{M-j} \frac{N^{N-M+j} (M-j)!}{N!} & j = 1, \dots, M - N + 1 \end{cases}$$

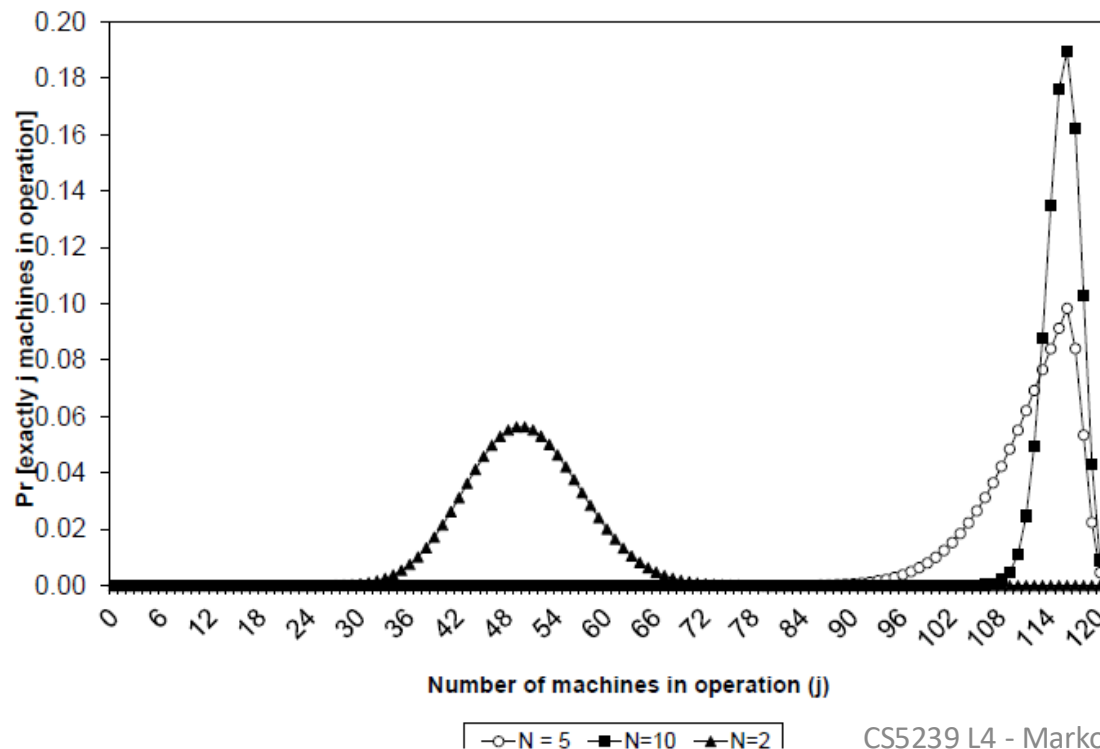
where

$$p_0 = \left[\sum_{k=0}^N \left(\frac{\lambda}{\mu} \right)^k \binom{M}{k} + \sum_{k=N+1}^M \left(\frac{\lambda}{\mu} \right)^k \binom{M}{k} \frac{N^{N-k} k!}{N!} \right]^{-1}$$

Model Interpretation

1. What is the probability that exactly j ($j = 1, \dots, M$) machines are operational at any given time?

For $N = 2, 5, 10$, we have this figure



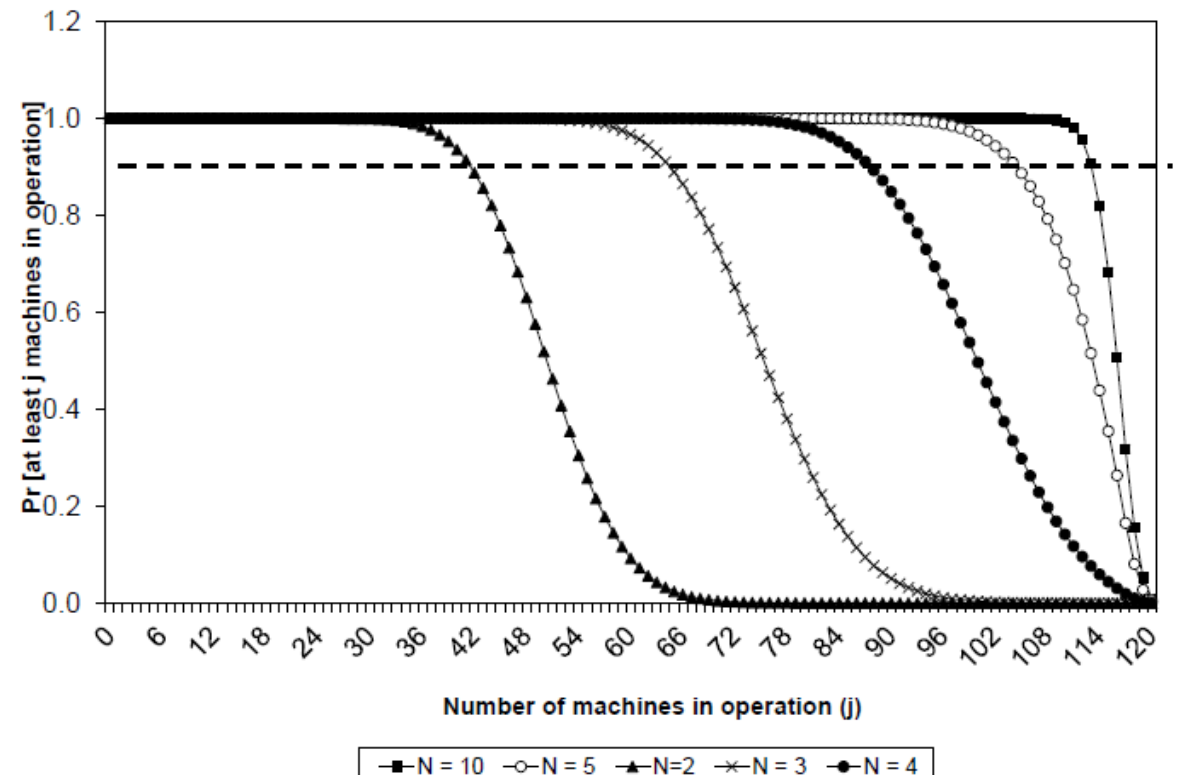
N	#machines in operation	probability
2	50	0.056
5	116	0.10
10	116	0.19

Model Interpretation

2. What is the probability that at least j ($j = 1, \dots, M$) machines are operational at any given time?

- The probability is $P_j = \sum_{i=j}^M p_{M-i}$

For $j = 1, 2, \dots, 120$ and for $N = 2, 3, 4, 5$ and 10



Model Interpretation

4. What is the effect of the size of the repair team on the mean time to repair (MTTR) a machine? What is the effect of the size of the repair team on the percentage of machine that can be expected to be operational at any given time?

MTTR is calculated by

$$\text{MTTR} = \frac{M}{\sum_{k=0}^{M-1} (M - k) \lambda p_k} - 1/\lambda$$

then we can compute the number of repairman as in the following table

Model Interpretation

4. continued ...

	operational	failed		probability of operational
N	N_o	N_f	MTTR	$N_o/M(\%)$
1	25.0	95.0	1900.0	20.8
2	50.0	70.0	700.0	41.7
3	75.0	45.0	300.0	62.5
4	99.2	20.8	104.8	82.7
5	111.5	8.5	38.1	92.9
6	114.3	5.7	25.1	95.2
7	115.0	5.0	21.7	95.8
8	115.3	4.7	20.6	96.0
9	115.3	4.7	20.2	96.1
10	115.4	4.6	20.1	96.1
120	115.4	4.6	20	96.2

Model Interpretation

5. What is the effect of the average time it takes a member of the repair team to fix a machine (i.e., their skill level) on the overall MTTR (i.e., includes the time waiting to repair)? How does a repair person's skill level affect the percentage of operational machine?

vary μ to fix repair
time 10-25 mins

Avg. repair time	μ	N_o	N_f	MTTR	$N_o/M(\%)$
10	0.100	117.6	2.4	10.4	98.0
12	0.083	117.0	3.0	12.9	97.5
15	0.067	115.8	4.2	18.1	96.5
18	0.056	113.8	6.2	27.2	94.8
20	0.050	111.5	8.5	38.1	92.9
25	0.040	99.1	20.9	105.5	82.6

Summary

- Analysis of single queue systems
- State-transition diagram and balance equations
- Example that shows how to solve more complex problems using Markov model

References:

- [Jain] – chapter 31
- Performance by Design, D.A. Menasce et al., Prentice-Hall – chapters 10 and 7 (read softcopy @ Google books)